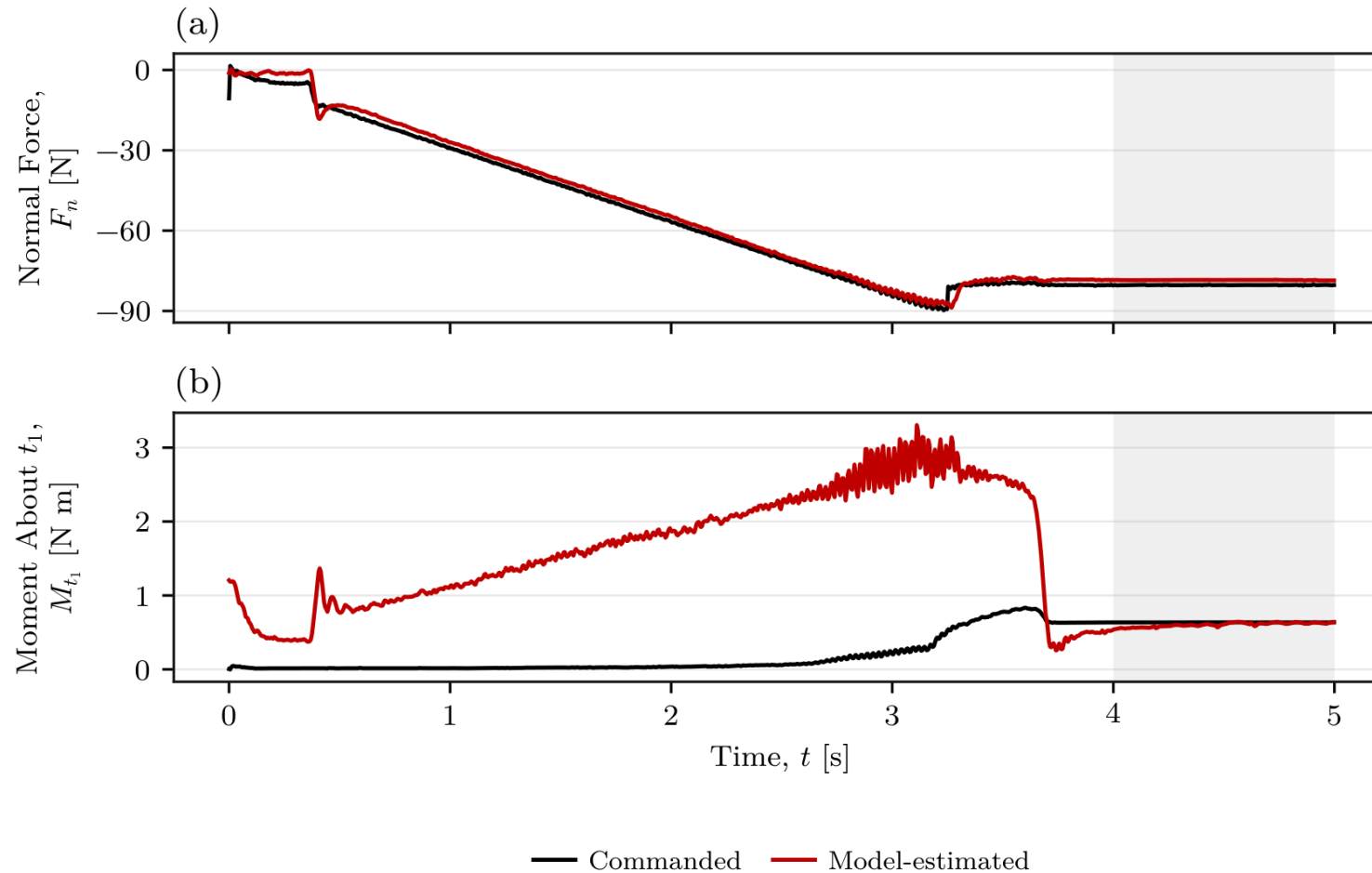
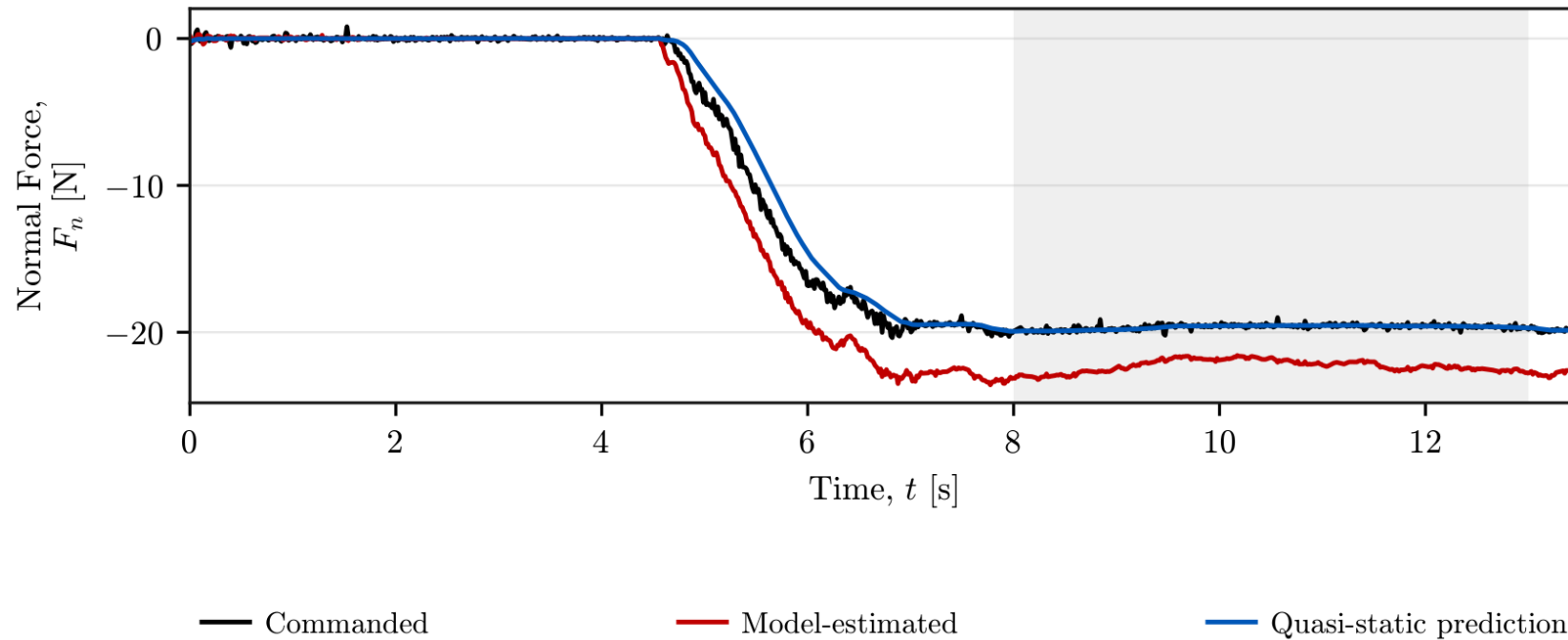


Commanded and estimated wrench



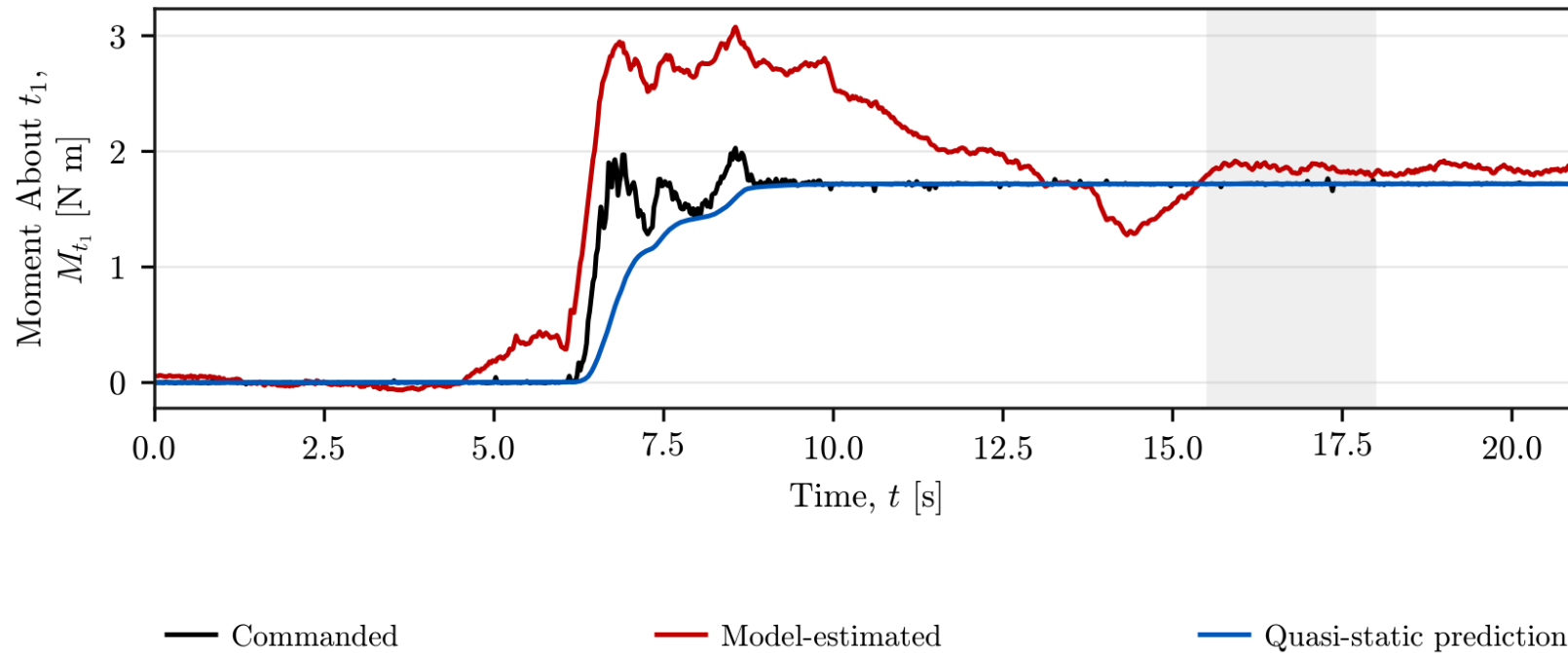
- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%
- Supplementary plausibility evidence

$$F_{n,qs} = K_{p,n} n_s^\top (e_p - e_{p,0})$$



- Spring vs command: 0.05% difference
- Estimate vs command: 13.50% difference

$$M_{t_1,qs} = K_{R,t_1} t_1^\top (e_R - e_{R,0})$$



- Spring and commanded mean: 1.719 N m
- Model-estimated mean: 1.852 N m

The EE frame is attached to the controlled tool centre point (TCP).

Position

$$p_{EE} \in \mathbb{R}^3$$

3 translational degrees of freedom
TCP position in the robot base frame

Orientation

$$R_{EE} \in SO(3)$$

3 rotational degrees of freedom
EE rotation relative to the robot base

Measured pose

$$T_{EE} = \begin{bmatrix} R_{EE} & p_{EE} \\ 0^\top & 1 \end{bmatrix}$$

Obtained directly from the robot state for position and orientation feedback.

7 joint coordinates control a 6-DOF tool pose: 3 position + 3 orientation.

$$\dim(\ker J) = 7 - \text{rank}(J) = 7 - 6 = 1$$

At a nonsingular configuration: one independent redundant motion direction.

This freedom is a coordinated motion of the joints, not one designated joint.

$$J(q) \dot{q}_{\text{null}} = 0, \quad J \in \mathbb{R}^{6 \times 7}$$

J maps joint velocity to tool velocity: null-space motion gives zero tool velocity.

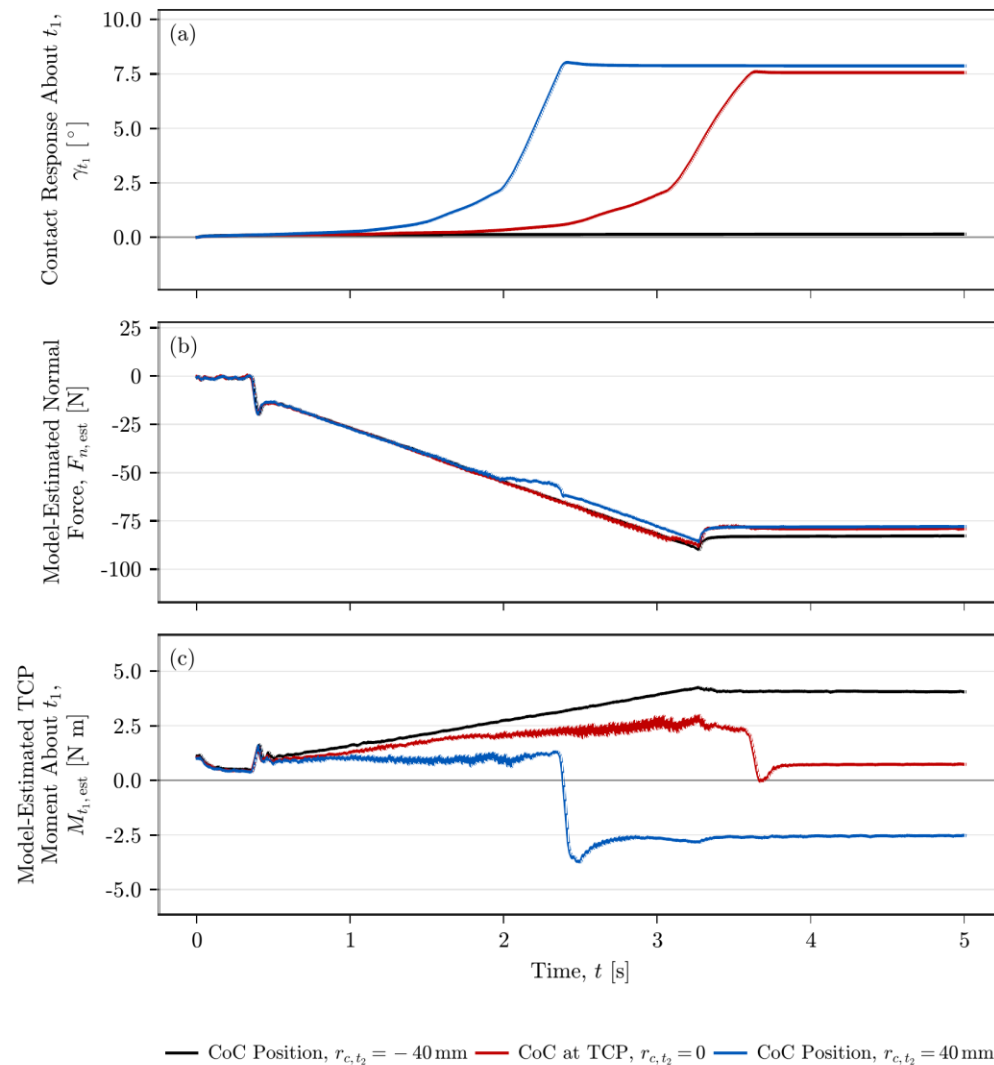
Two torque components control this internal motion:

$$\tau_{\text{null}} = \tau_d + \tau_{\sigma}$$

Damping torque: counters null-space joint motion.

Conditioning torque: steers towards a configuration with better motion capability.

Contact response, normal force and moment



Representative traces Positive entry tilt

- Similar final normal force
- Moment reverses with CoC displacement
- Force and moment: model-based robot estimates